Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d-step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1-\lambda_c)\, \boldsymbol{\delta x[k]} - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k] f_T[k]$$

d-step measurement delay:

$$\boldsymbol{\delta x[k]} = \delta x[k-d] + \sum_{i=1}^{d} \Delta x_d[k-i] - \sum_{i=1}^{d} a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^{d} \delta x_D[k-i]$$

Ideal control effort (with d-step measurement delay):

$$\bar{\bar{f}}_{dx}[k] = a_x^{-1}[k]\Big\{ \underbrace{\Big[\Delta x_d[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} \Delta x_d[k-i]\Big]}_{\Delta x_d^d[k]} + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big]$$
$$- \underbrace{\Big[\delta x_D[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} \delta x_D[k-i]\Big]}_{\delta x_D^d[k]} \Big\}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\Big\{ a_x[k] f_T[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} a_x[k-i] f_T[k-i] \Big\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k]\big[f_{dx}[k] - \bar{\bar{f}}_{dx}[k]\big] - \delta x_T^d[k]$$

$$a_x[k]\,\bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big] - \delta x_D^d[k]$$

$$a_x[k]\, f_{dx}[k] = \big(1 + e_{a_x}[k]\,\hat{a}_x^{-1}[k]\big)\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\big(\delta x[k-d] + n_x[k]\big) - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}$$

$$\begin{aligned} a_x[k]\, f_{dx}[k] - a_x[k]\,\bar{\bar{f}}_{dx}[k] = e_{a_x}[k]\, f_{dx}[k] + (1-\lambda_c) n_x[k] \\ + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \big(a_x[k-i] - \hat{a}_x[k-i]\big) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \qquad a_{x_{ram}}[k] \approx a'_x[k]\, x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= a_{x_{det}}[k+1] + a_{x_{ram}}[k+1] = \underbrace{\big\{a_{x_{det}}[k] + a_{x_{ram}}[k]\big\}}_{a_x[k]} + \Delta a_x[k] + \underbrace{\big\{a_{x_{ram}}[k+1] - a_{x_{ram}}[k]\big\}}_{\Delta a_{ram}[k]} \\ &= a_x[k] + \Delta a_x[k] + \Delta a_{ram}[k] \end{aligned}$$

$\Delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a'_x[k]\,\Delta h_d[k]$ (from p_d), $\qquad \Delta a_{ram}[k] = a'_x[k]\,\Delta x_{ram}[k]$

$\Delta h_d[k] = h_d[k+1] - h_d[k]$ (desired-trajectory height increment).

Gain dynamics — $a_{x_{ram}}$ is first-order autoregressive (AR(1))

Assumption ($x = x_{det} + x_{ram}$, perfect tracking $x_{det} = x_d$): $\delta x = x_d - x = -x_{ram}$ (zero-mean), $x_{ram} = -\delta x$; ideal closed loop ($\hat{a}_x = a_x$, $\delta x_D^d \equiv 0$) gives $\delta x[k+1] = \lambda_c\,\delta x[k] - \varepsilon[k]$ (ε: closed-loop residual, defined in the boxed motion error below).

$$a_{x_{ram}}[k+1] = -a'_x\,\delta x[k+1] = -a'_x\big(\lambda_c\,\delta x[k] - \varepsilon[k]\big) = \lambda_c\, a_{x_{ram}}[k] + a'_x\,\varepsilon[k]$$

$$\begin{aligned} a_x[k+1] &= \big(a_{x_{det}}[k] + \Delta a_x[k]\big) + \big(\lambda_c\, a_{x_{ram}}[k] + a'_x\,\varepsilon[k]\big) \\ &= \big(a_{x_{det}}[k] + \Delta a_x[k]\big) + \lambda_c\big(a_x[k] - a_{x_{det}}[k]\big) + a'_x\,\varepsilon[k] \\ &= \underbrace{\lambda_c\, a_x[k]}_{\mathbf{F}_e(4,4)=\lambda_c} + \underbrace{(1-\lambda_c)\, a_{x_{det}}[k] + \Delta a_x[k]}_{\text{known input}} + \underbrace{a'_x\,\varepsilon[k]}_{q_4[k]} \end{aligned}$$

$$\boxed{\mathbf{F}_e(4,4) = \lambda_c}$$

Parameter estimator (predict reverts to $a_{x_{det}}$): $\quad \hat{a}_x[k+1] = \lambda_c\,\hat{a}_x[k] + (1-\lambda_c)\,a_{x_{det}}[k] + \Delta a_x[k]$

Parameter estimation error: $\quad e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$

Estimation error dynamics: $\quad e_{a_x}[k+1] = \lambda_c\, e_{a_x}[k] + a'_x\,\varepsilon[k]$

$$a_x[k-i] - \hat{a}_x[k-i] = e_{a_x}[k-i] \overset{(\dagger)}{\approx} e_{a_x}[k] - \sum_{j=1}^{i} \Delta a_{ram}[k-j]$$

(†) feedforward form $e_{a_x}[k{+}1]{=}e_{a_x}[k]{+}\Delta a_{ram}$ is exact; the AR(1) predictor gives $e_{a_x}[k{+}1]{=}\lambda_c e_{a_x}{+}a'_x\varepsilon$, altering the expansion by $O\big((1{-}\lambda_c)e_{a_x}\big)$ which enters only $\delta x^d_{\delta af}$ ($\ll \delta x^d_T$ in ε).

$$\begin{aligned} a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{f}}_{dx}[k] \approx e_{a_x}[k]\, & \underbrace{\Big\{ f_{dx}[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} f_{dx}[k-i] \Big\}}_{F_{dx}[k]} \\ & - \underbrace{\Big\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} (d+1-i)\, f_{dx}[k-i]\, \Delta a_{ram}[k-i] \Big\}}_{\delta x^d_{\delta af}[k]} \\ & + \delta x^d_D[k] + (1-\lambda_c) n_x[k] \end{aligned}$$

$$\boxed{\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x^d_D[k] - \underbrace{\Big\{ \delta x^d_T[k] - \delta x^d_{\delta af}[k] + (1-\lambda_c) n_x[k] \Big\}}_{\varepsilon[k]}}$$

State equation

($\delta x^d_D \equiv 0$: disturbance not modelled)

$$\begin{aligned} \delta x_1[k+1] &= \delta x_2[k] \\ \delta x_2[k+1] &= \delta x_3[k] \\ \delta x_3[k+1] &= \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k] \\ a_x[k+1] &= \lambda_c\, a_x[k] + (1-\lambda_c)\, a_{x_{det}}[k] + \Delta a_x[k] + a'_x\, \varepsilon[k] \end{aligned}$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^{d} \Delta a_x[k-i] + \underbrace{\Big\{ n_a[k] - \sum_{i=1}^{d} \Delta a_{ram}[k-i] \Big\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^{d} \Delta a_x[k-i] \end{bmatrix} \text{ (known)}, \qquad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

Past-error expansion under AR(1) (†)

Two predictors, two e_{a_x} recursions (predict-only):

$$\begin{aligned}\text{feedforward:}\quad & e_{a_x}[k+1] = e_{a_x}[k] + \Delta a_{ram}[k] \;\Rightarrow\; e_{a_x}[k-i] = e_{a_x}[k] - \textstyle\sum_{j=1}^{i} \Delta a_{ram}[k-j] \quad \text{(exact)}\\ \text{AR(1):}\quad & e_{a_x}[k+1] = \lambda_c\, e_{a_x}[k] + a_x'\varepsilon[k] \;\Rightarrow\; e_{a_x}[k-i] = \lambda_c^{-i}\big(e_{a_x}[k] - \textstyle\sum_{j=1}^{i} \lambda_c^{j-1} a_x'\varepsilon[k-j]\big)\end{aligned}$$

Difference ($\Delta_i := e_{a_x}^{\mathrm{AR}(1)}[k-i] - e_{a_x}^{\mathrm{ff}}[k-i]$):

$$\Delta_i = \underbrace{(\lambda_c^{-i} - 1)\, e_{a_x}[k]}_{\text{reverting}} - \lambda_c^{-i} \sum_{j=1}^{i} \lambda_c^{j-1} a_x'\varepsilon[k-j] + \sum_{j=1}^{i} \Delta a_{ram}[k-j]$$

Leading order (small i, $\lambda_c^{-i} \approx 1 + i(1-\lambda_c)$):

$$(\lambda_c^{-i} - 1)\, e_{a_x}[k] \approx i(1-\lambda_c)\, e_{a_x}[k] = O\big((1-\lambda_c)\, e_{a_x}\big)$$

$\Rightarrow$ the feedforward expansion errs by $O((1-\lambda_c)e_{a_x})$, entering ε only through $\delta x^d_{\delta af}$ ($\ll \delta x^d_T$); numerically Q_{33}/R_{22} with $\mathrm{Var}(\Delta a_{ram})$ vs q_4 leaves $\hat{a}_x$ unchanged.

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta\hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \Big(\hat{a}_x[k] - \textstyle\sum_{i=1}^{d} \Delta a_x[k-i]\Big) \\ &= \Big(a_x[k] - \textstyle\sum_{i=1}^{d} \Delta a_x[k-i] + r_2[k]\Big) - \hat{a}_x[k] + \textstyle\sum_{i=1}^{d} \Delta a_x[k-i] \\ &= e_{a_x}[k] + \underbrace{\Big\{n_a[k] - \textstyle\sum_{i=1}^{d} \Delta a_{ram}[k-i]\Big\}}_{r_2[k]} \end{aligned}$$

$$\begin{aligned} \delta\hat{x}_1[k+1] &= \delta\hat{x}_2[k] + \ell_{11}[k]\, e_{y_1}[k] + \ell_{12}[k]\, e_{y_2}[k] \\ \delta\hat{x}_2[k+1] &= \delta\hat{x}_3[k] + \ell_{21}[k]\, e_{y_1}[k] + \ell_{22}[k]\, e_{y_2}[k] \\ \delta\hat{x}_3[k+1] &= \lambda_c \cdot \delta\hat{x}_3[k] + \ell_{31}[k]\, e_{y_1}[k] + \ell_{32}[k]\, e_{y_2}[k] \\ \hat{a}_x[k+1] &= \lambda_c\, \hat{a}_x[k] + (1-\lambda_c)\, a_{x_{det}}[k] + \Delta a_x[k] + \ell_{41}[k]\, e_{y_1}[k] + \ell_{42}[k]\, e_{y_2}[k] \end{aligned}$$

Error dynamics

$$\begin{aligned} e_{\delta x_1}[k+1] &= e_{\delta x_2}[k] - \ell_{11}[k]\, e_{\delta x_1}[k] - \ell_{12}[k]\, e_{a_x}[k] \\ e_{\delta x_2}[k+1] &= e_{\delta x_3}[k] - \ell_{21}[k]\, e_{\delta x_1}[k] - \ell_{22}[k]\, e_{a_x}[k] \\ e_{\delta x_3}[k+1] &= \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k]\, e_{\delta x_1}[k] - \ell_{32}[k]\, e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]} \\ e_{a_x}[k+1] &= \lambda_c\, e_{a_x}[k] - \ell_{41}[k]\, e_{\delta x_1}[k] - \ell_{42}[k]\, e_{a_x}[k] + \underbrace{a_x'\, \varepsilon[k]}_{q_4[k]} \end{aligned}$$

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & \lambda_c \end{bmatrix}}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^{d} f_{dx}[k-i]$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k]\,\mathbf{r}[k]$$

Process / measurement noise variances

Assumption ($x = x_{det} + x_{ram}$, perfect tracking): $x_{ram} = -\delta x$ (zero-mean), $a'_x := \dfrac{da_x}{dh} = -\dfrac{a_x K_h}{R}$, $K_h = \dfrac{1}{c}\dfrac{dc}{d\bar{h}}$ (wall sensitivity).

Closed-loop tracking variance (thermal-driven, $\hat{a}_x = a_x$):

$$\mathrm{Var}(x_{ram}) = \mathrm{Var}(\delta x) = C_{\delta x}\,\sigma^2_{\delta h}, \qquad C_{\delta x} = 2 + \frac{1}{1-\lambda_c^2}, \qquad \sigma^2_{\delta h} := 4k_B T\,a_x$$

(i) True increment $\Delta a_{ram} = a'_x\,\Delta x_{ram}$ ($\rho_1 = \mathrm{Cov}(x_{ram}[k{+}1], x_{ram}[k])/\mathrm{Var}(x_{ram})$, lag-1 autocorr):

$$\mathrm{Var}(\Delta x_{ram}) = 2(1-\rho_1)\,\mathrm{Var}(x_{ram}), \qquad \rho_1 = \lambda_c + \frac{2(1-\lambda_c)}{C_{\delta x}} \;\Rightarrow\; 1-\rho_1 = \frac{1}{(1+\lambda_c)\,C_{\delta x}}$$

$$\boxed{\mathrm{Var}(\Delta a_{ram}) = a'^2_x \cdot \underbrace{2(1-\rho_1)\,C_{\delta x}}_{=2/(1+\lambda_c)}\,\sigma^2_{\delta h} = \frac{2}{1+\lambda_c}\,a'^2_x\,\sigma^2_{\delta h} \qquad (\to Q_{33}^{\text{randgain}},\; R_{22}^{\text{delay}})}$$

(ii) AR(1) white driver $w_4 = a'_x\,\varepsilon$ ($\mathrm{Var}(\varepsilon) = (1-\lambda_c^2)\,\mathrm{Var}(\delta x)$ since δx is AR(1)):

$$\boxed{q_4 = a'^2_x\,\mathrm{Var}(\varepsilon) = (1-\lambda_c^2)\,C_{\delta x}\,a'^2_x\,\sigma^2_{\delta h} = (3-2\lambda_c^2)\,a'^2_x\,\sigma^2_{\delta h} \qquad (\to Q_{44})}$$

Self-consistency (AR(1) steady-state variance = true gain variance):

$$\frac{q_4}{1-\lambda_c^2} = C_{\delta x}\,a'^2_x\,\sigma^2_{\delta h} = \mathrm{Var}(a_{x_{ram}})\ \checkmark$$

Impl.: K_h, $\sigma^2_{\delta h} = 4k_B T\,a_\perp$ evaluated at $\bar{h}_{\text{meas}}$ (p_m); gain $a_x \to \hat{a}_x$ (estimate); reversion target $a_{x_{det}}$ from p_d.

Original random-walk Q_{44} — the mismatch

Original model (RW, $\mathbf{F}_e(4,4)=1$): predict mean and Q_{44} are mutually consistent,

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \Delta a_x[k], \qquad Q_{44} = \mathrm{Var}(\Delta a_{ram}).$$

True increment (from $a_{x_{ram}}[k+1] = \lambda_c\, a_{x_{ram}}[k] + a'_x \varepsilon[k]$):

$$\Delta a_{ram}[k] = a_{x_{ram}}[k+1] - a_{x_{ram}}[k] = \underbrace{(\lambda_c - 1)\, a_{x_{ram}}[k]}_{\text{reversion (spring)}} + \underbrace{a'_x\, \varepsilon[k]}_{\text{white}}$$

Mismatch: RW puts the *whole* Δa_{ram} into Q_{44} — it treats the deterministic reversion $(\lambda_c - 1)a_{x_{ram}}$ as white noise; $\mathbf{F}_e(4,4) = 1$ keeps no spring, so the modelled gain variance is unbounded while the true gain is bounded:

$$\text{RW: } P_a[k+1] = P_a[k] + Q_{44} \ \nearrow \infty \qquad \text{AR(1): } P_a[k+1] = \lambda_c^2 P_a[k] + q_4 \to \tfrac{q_4}{1-\lambda_c^2}$$

Fix: move the reversion into the dynamics ($\mathbf{F}_e(4,4) = \lambda_c$, predict reverts to $a_{x_{det}}$), keep only the white driver in Q_{44}:

$$\underbrace{\mathrm{Var}(\Delta a_{ram}) = \tfrac{2}{1+\lambda_c}\, a_x'^2 \sigma_{\delta h}^2}_{\text{RW } Q_{44}} \ \longrightarrow\ \underbrace{q_4 = (3 - 2\lambda_c^2)\, a_x'^2 \sigma_{\delta h}^2}_{\text{AR(1) } Q_{44}}$$

Effect (1 Hz, near wall, z-axis): $P_a/\mathrm{Var}(a_{x_{ram}}):\ 19.67 \to 0.78, \quad |\hat{a}_z - a_z|:\ 9.4\% \to 0.4\%.$